Chapter-01

Ques-01:- What is health economics? Explain the relevance of health economics.

Answer:-

Health Economics:-

Health economics is a branch of economics concerned with issues related to efficiency, effectiveness, value and behaviour in the production and consumption of health and health care. Health economics studies how resources are allocated to and within the health economy. The production of health care and its distribution across population fall within this defination. Kenneth Arrow is the father of health economics.

In broad terms, health economists study the functioning of the health care systems as well as health affecting behaviors such as smoking. Health economics is a sub-discipline of economics that is concerned with the efficient allocation of health care resources.

The Relevance of Health Economics:-

The study of health economics is important and interesting in three related ways:-

(1) The size of the contribution of the health sector to the overall economy.

(2) The national policy concerns resulting from the importance many people attach to the economic problems ~~as~~ they face in pursuing and maintaining their health, and

(3) The many health issues that have a substantial economic element.

Features of Economic Analysis:-

Many distinctive features of economics might be identified, but we emphasize four:-

(i) Scarcity of social resources
(ii) Assumption of rational decision making
(iii) Concept of marginal analysis
(iv) Use of economic model

(i) Scarcity of Resources:-

Economic analysis is based on the premise that individuals must give up some of one resource in order to get some of another. At the national level, this means that increasing shares of GDP going to health care ultimately imply decreasing shares going for other uses. Economists view time as the ultimate scarce resource. Individuals sell their time for wages, and many individuals will refuse overtime work even if offered more than their normal wage rate. Similarly many will pass up free health care because the travel and waiting time costs are too high.

(ii) Rational decision making:-

Economists typically approach problems of human economic behaviour by assuming that the decision maker is a rational being. Rationality is effectively defined as "making choices that best further one's own ends given one's resource constraints." Some behaviours may appear irrational.

(iii) Marginal Analysis:-

Mainstream economic analyses feature reasoning at the margin. To make an appropriate choice, decision makers must understand the cost as well as the benefit of the next or marginal unit.

(iv) Use of Models:-

Finally economics characteristically develops models to depict its subject matter. The models may be described in words, graphs or mathematics.

Ques-02:- Does economics apply to health and health care? Explain.

Answer:-

If economics studies how scarce resources are used to produce goods and services, and ~~to~~ then how these goods and services are distributed, then clearly economics applies ~~to~~ to health and health care. Certainly health care resources are scarce; in fact, their cost concerns most people. There is no question that health care is produced and distributed.

However, much ~~of~~ ~~the~~ health care simply does not fit this image. A considerable amount of health care is elective, meaning that patients have and will perceive some choice over whether and when to have the diagonostics or treatment involved. Much health care is even routine, involving problems such as ~~to~~ upper respiratory infections, back pain, and diagonostic checkups. The patient often has prior experience with ~~this~~ these concerns. Furthermore even in a real emergency, consumers have agents to make or help making decisions on their behalf. Traditionally physicians have served as agents and more recently, care managers have also entered the process. Thus, rational choices can be made.

Does Price Matter? :-

Many have argued that health care is so different from other goods that consumers do not respond to financial incentives. Following figure examines the use of ambulatory mental health and medical care where amounts of health care consumed are measured along the horizontal axis. These amounts are scaled in percentage ~~form~~ terms from zero to 100 percent, where 100 percent reflects the average level of care consumed by the group that used the most care on average.

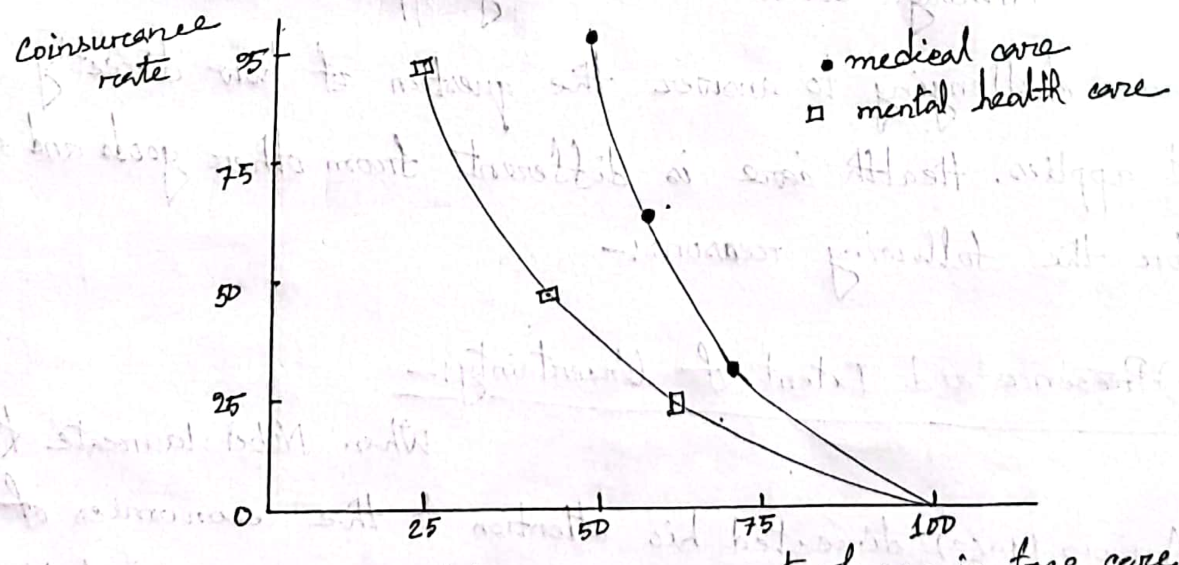


Fig: Demand response of ambulatory mental and medical care

This group is the group with free care. The vertical axis measures the economic incentives as indicated by the coinsurance rate - the percentage of the bill paid out directly by the consumer. Thus a higher coinsurance rate reflects a higher price to the consumer. The figure shows that people are consuming more care as the care becomes less costly to the consumer in terms of dollars paid out of pocket. More importantly, the curve demonstrates that economic incentives do matter. Those facing higher prices demand less care.

Ques-03:- Why is health care is different from other goods and services. Explain.

Answer:-

Although economics certainly applies to health care, it is more challenging to answer the question of how directly and it applies. Health care is different from other goods and services for the following reasons:-

(i) Presence and Extent of Uncertainty:-

When Nobel laureate Kenneth Arrow (1963) directed his attention to the economics of health, he helped establish health economics as a field. He stressed the prevelance of uncertainty in health care, on both the demand side and the supply side. Consumers are uncertain of their health states and need for health care in any coming period. This means that the demand for health care is irregular in nature from the individual's perspective; likewise the demand facing a health care firm is irregular.

Uncertainty is also prevalent on the supply side. Standard economic analysis often assumes that products and the pleasures that they bring, are well understood by the purchasers. The purchase of steak, milk, new clothes etc prodives expected well-being that is easily known and understood. In contrast, several cases of product uncertainty exist in the health field. Consumers often do not know the expected outcomes of various treatments without physician's advice and in many cases physicians themselves cannot predict the outcomes of treatments with certainty.

(ii) Prominence of Insurance:-

Consumers purchase insurance to gaurd against this uncertainty and risk. Because of health insurance, neither most Americans nor citizens of other countries pay directly for the full cost of their health care.

Suppose, in a specific year almost 52 percent of all personal health care expenditures were paid out of pocket meaning that 48 percent was paid by the third party payers. Insurance change the demand for health care and it potentially also changes the incentive of providers.

(iii) Problems of Information:-

Uncertainty can in part be attributed to lack of information. Sometimes information is unavailable to all parties concerned. For example, neither gynecologists nor their patients may recognize the early stages of cervical cancer without Pap smears. At other times, the information in question is known to some parties, but not to all, and it is the asymmetry of information that is problematic.

The problems of information means that careful economic analysis must modify their methods. Standard analysis often assume that consumers have the necessary knowledge about the ods.

Consumers may not know which physicians or hospitals are good, capable or even completent. They may not know whether they are ill or what should be done if they are. This lack of information often makes a consumer dependent on the provider. The provider offers both the information and the service leading to the possibility of conflicting interests.

(iv) Large Role of Nonprofit Firms:-

Economists often assume that firms maximize profits. Yet many health care providers, including many hospitals, insurers, and nursing homes, have nonprofit status. The economists must analyze the establishment and perpetuation of nonprofit institutions and understand the differences in their behaviours from for-profit firms. The public and the larger medical community are aware of the major hospitals as centers of health care, teaching, and research.

(v) Restrictions on Competitions:-

The health sector has developed many practices that effectively restrict competition. These practices include ~~licens~~ licensure requirements for providers, restrictions on provider advertising, and standards of ethical behaviour that enjoin providers from competing with each other. Regulation to promote quality or curb costs also reduces the freedom of choice of providers and may influence competition.

(vi) Role of Equity and Need:-

Poor health of another human being often evokes a feeling of concern that distinguishes health care from many other goods and services. Many advocates express this feeling by saying that people ought to get the health care they need regardless of whether they can afford it.

(vii) Government Subsidies and Public Provision:-

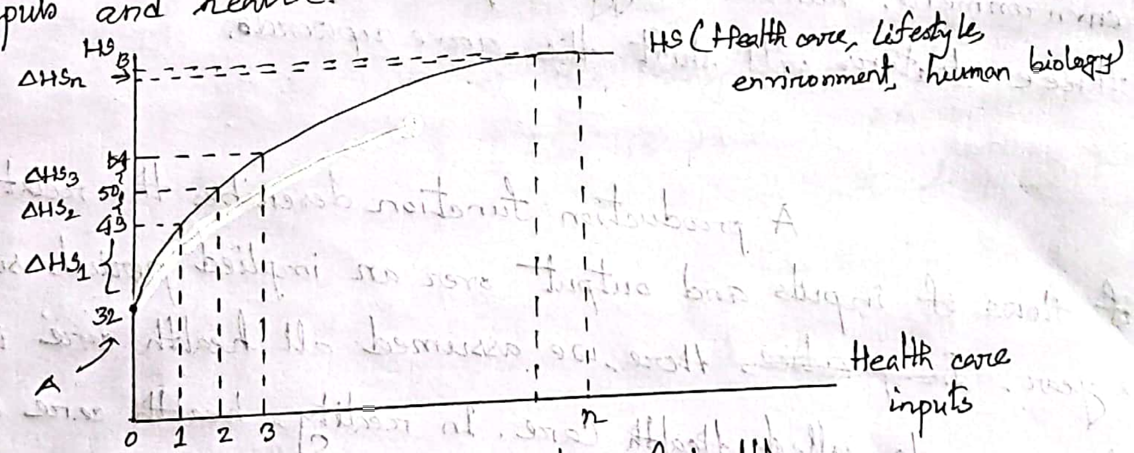
In most countries, the government plays a major role in the provision or financing of health services.

Ques-01:- Define health production function. Would the health production function eventually bend downward? Explain in support of your answer.

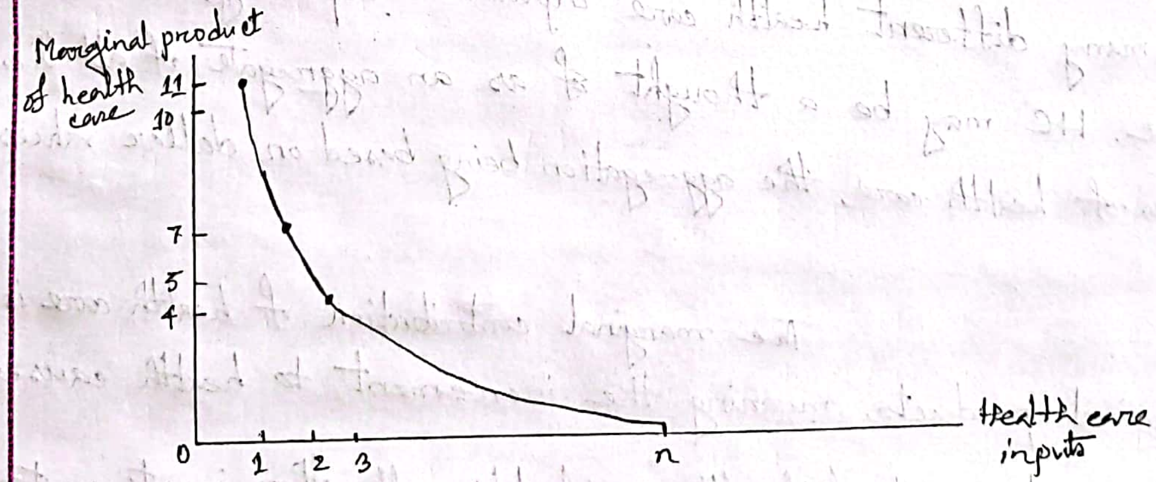
Answer:

Health Production Function:-

A production function summarizes the relationship between inputs and outputs. The study of the production of health function requires that we inquire about the relationship between health inputs and health.



A. Production function of health



B. Marginal product of health care

Fig: Production of Health

The marginal contribution of health care is its marginal ~~contribution of health~~ product, meaning the increment to health caused by one extra unit of health care, holding all other inputs constant. The distinction between total and marginal contribution of health care is illustrated in the figure. It exhibits a theoretical health status production function for the population. Health status is shown here to be an increasing function of health care. Here, health status, HS = health care, lifestyle, environment, human biology. Improvements in any of these latter three factors will shift the curve upwards.

A production function describes the relationship of flows of inputs and output over an implied period, such as a year. ~~In practice,~~ Here, we assumed all health care inputs into one scale called Health Care. In reality, health care consits of many different health care inputs. Conceptually, the health care, HC may be a thought of as an aggregate of all these types of health care, the aggregation being based on dollar values.

The marginal contribution of health care is its marginal products, meaning the increment to health caused by one extra unit of health care, holding all other inputs constant.

Increasing Health Care from zero to one unit in the figure(A) improves health status by ΔHS_1, the first unit's marginal product. Numerically, the first unit of Health Care has increased the health status index from 32 to 43, $\Delta HS_1 = 11$ Health status units. The next unit of medical care delivers a marginal product of $\Delta HS_2 = 7$, and so on.

These marginal products are diminishing in size, illustrating the law of diminishing marginal returns. If society employs a total of n units of health care, then the total contribution of Health Care is the sum of the marginal products of each of the n units. This total contribution is shown as AB, may be substantial. However, the marginal product of the nth unit of Medical Care is ΔHS_n, an it is small. In fact, we are nearly on the 'flat of the curve'. Marginal product is graphed on figure(B).

Downward Bending of Health Production Function:- We have drawn in the figure the production function as a rising curve that flattens out at higher levels of health care but never bends downward. But the health production function eventually bends downward. This is a logical possibility under at least two scenarios:-

(i) Iatrogenic (provider caused) disease is an inevitable by-product of many medical interventions. For example, each surgery has its risk.

(ii) Combinations of drugs may have unforeseen and adverse interactions.

Many medical scientists have pressed the case that much ~~more~~ medical care as often practiced has only weak scientific basis, making iatrogenesis a real ~~problem~~ probability. Illich argued that medicalization would lead to less personal effort to preserve health and less personal determination to preserverere; the result becomes a decline in the health of the population and thus a negative margenal product for medical care.

Ques-02:- Explain the two different theories of schooling.

Answer:-

First, Michael Grossman's theory of demand entails a central role for education. Grossman contends that better educated person tends to be economically more efficient producers of health status. Better educated people understand the technology or have the knowledge-how needed to stay healthy. They also know better how to use medical and other market inputs and their own time to produce health. Under this view, the marginal transfer of resources makes good sense.

Much of the available research concentrates formal schooling. other health economists have put forward hypothesis under which schooling and health status are correlated only because they are both related to one or more other factors.

Second, Victor Fuchs has suggested that people who seek out additional education tend to be those ~~who~~ with lower discount rates. A decision maker with a high discount rate will tend to prefer projects with immediate payoffs versus long-term projects. People with a lower discount rate tend to be those who value the long-term gains more. Now consider individuals facing a possible investment in education. Because education requires current costs to gain distant payoffs individuals with relatively low discount rates will be more likely to invest in education and in health as well.

Ques-01:- Explain home good function and health investment function in the light of Grossman model.

Answer:-

An increment to capital stock, such as health, is called an investment. Consider a consumer Edward, who buys market inputs (e.g. medical care, food, clothing) and combines them with his own time to produce the stock of health capital that produces services that increases his utility. Edward uses market inputs and personal time both to invest in his stock of health and to produce other things that he likes. During each period, Edward produces an investment in health, I. Thus,

Health investment, $I = I(M, T_H)$

Here, M = Market health inputs (providers' service, drugs)

T_H = Time spent improving health

This function indicates that, increased amount of M and T_H increase I.

There are also other items include virtually all other things that Edward does. They include time spent watching television, reading, playing with and teaching his children, preparing meals, baking bread or watching the sun set, a composite of other things people do with leisure time. We shall call this composite home good B.

Home good function, $B = B(X, T_B)$

Here, X = market purchased goods

T_B = Time for producing home goods

Home good (B) is produced with ~~time~~ home, T_B and market purchased goods, X. This function indicates that increased amounts of X and T_B increase B.

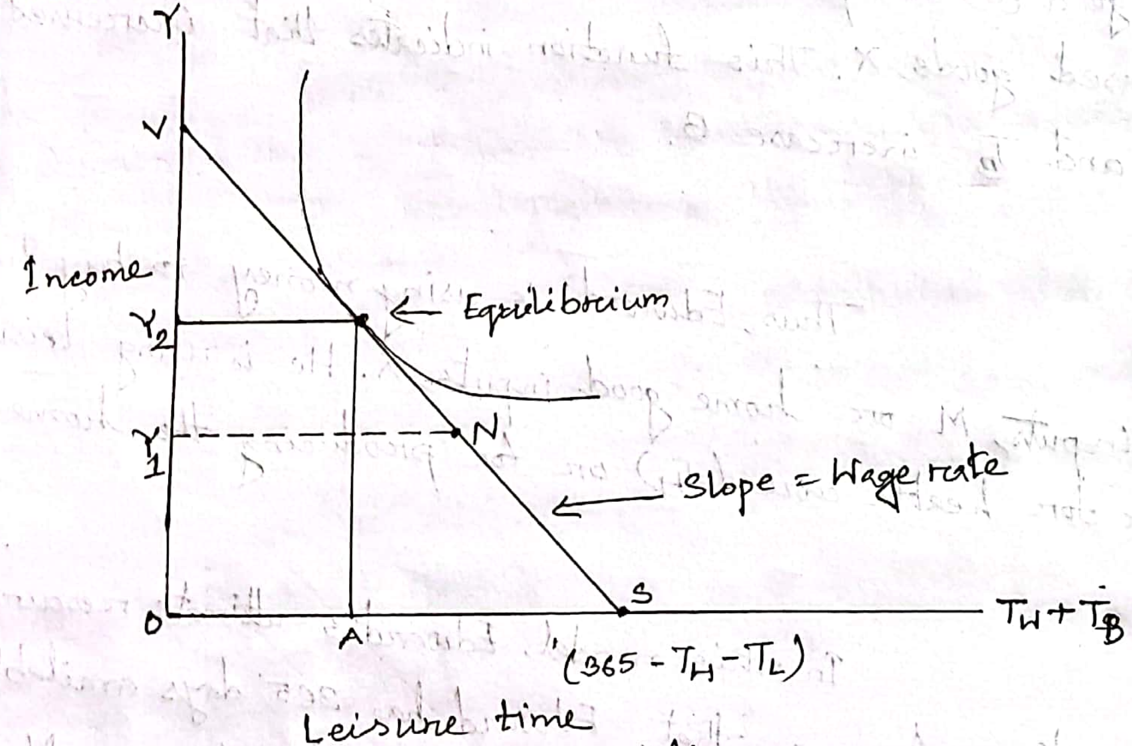
Thus, Edward is using money to buy health care inputs, M or home good inputs, X. He is using leisure time either for health care (T_H) or for producing the home good (T_B).

In this model, Edward's ultimate resource is his own time. Assume that, Edward has 365 days available in the year. To buy market goods such as medical care, M or other goods X, he must trade some of his time for income; that is he must work at a job. Call his time devoted to work T_W, we realize that some of his time during each year will be taken over by ill health or T_L. Thus we account for his total time in the following manner:

Total time $= T = 365$ days $= T_H$ (Improving health) $+ T_B$ (Producing home goods) $+ T_L$ (Lost to illness) $+ T_W$ (working).

Ques-02:- How would you explain the potential use of an individual's time by labor-Leisure trade off? Show the equilibrium of an individual's time labor- Leisure line and indifference curve.

Answer:-



Figure; Labor-Leisure Trade-off

The potential uses of a person's time are illustrated by the labor-Leisure trade-off. In the figure, the X axis represents Ed's work and leisure time. Suppose that he considers his spent creating health investment to be "health-improvement time" and that he calls T_B his Leisure. Assume that the number of days lost to ill health and the number of days spent on health-enhancing activity are fixed. Variables

T_L and T_H, refer to time lost and time spent on healthy activities, respectively. The maximum amount of time that he has available to use either for work, T_W, or leisure, T_B, is thus $365 - T_H - T_L$, so:

$$\text{Time available for work or Leisure} = 365 - T_H - T_L$$
$$= T_B + T_W$$

Leisure time, T_B, is measured toward the right while time spent at work, T_W, is measured toward the left. Figure shows that if Ed chooses leisure time, OA, then he has simultaneously chosen the amount of time at work indicated by AS.

Ed's total amount of time available for either work or Leisure is given by point S. If he chooses point S for the period, he would be choosing to spend all this available time in Leisure. The y-axis represents income, obtainted through work. This income will then purchase either market health goods or other market goods. Thus, if he chooses point S, he will not be able to purchase market goods because he has no wage income.

If Ed gives up one day of Leisure by spending that day at work, to point N, he will generate income equal to OY_1, which represents his daily wage. In economic terms, this quantity represents income divided by days worked - that is, the daily wage. The slope of the line VS depicting the labor-leisure trade-off reflects the wage rate. In equilibrium, Ed's trade-off of leisure and income is the same as the market trade-off, which is the wage rate. Here, he takes amount DA of leisure and trades amount AS of leisure for income, OY_2.

Ques-03:- How is it possible to earn more money and more leisure time together? Explain in the light of Grossman model.

Answer:-

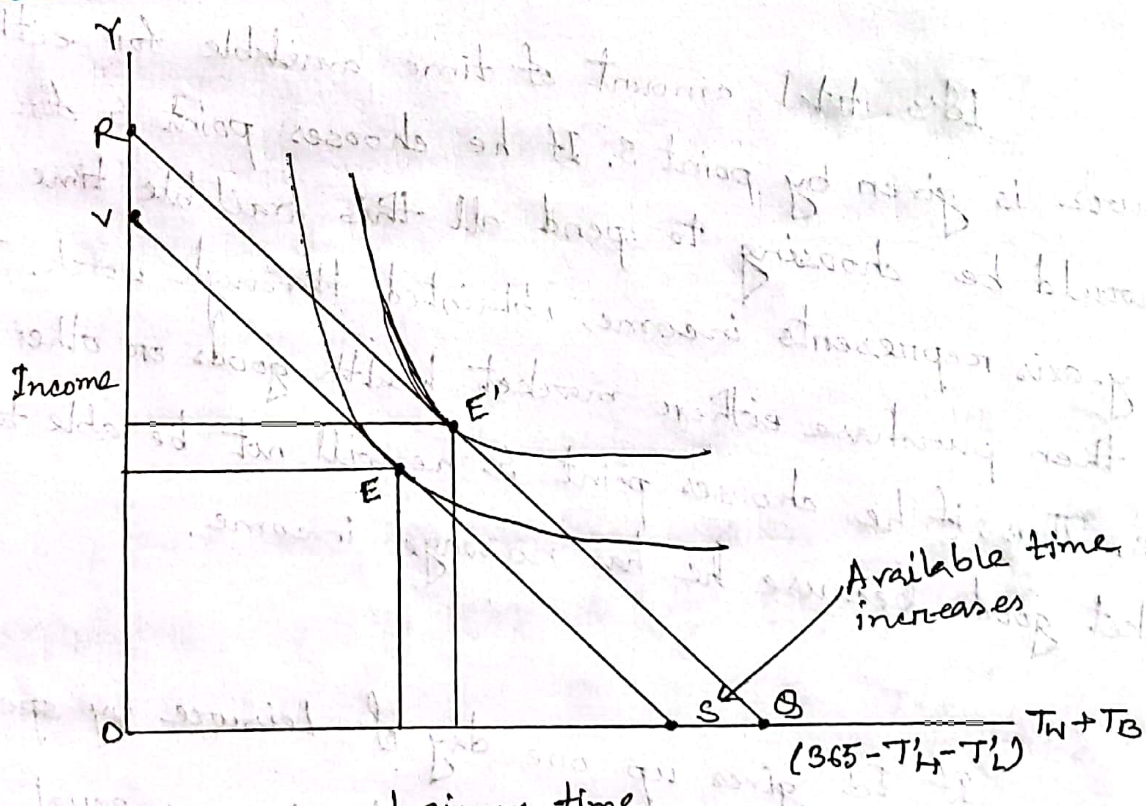


Figure: Increased amount of healthy time due to investment

An increase in activities to improve health, increased the time spent investing in health status. Suppose that time spent on health producing activities, T_H, is increased to T'_H. Correspondingly suppose that the number of days lost to ill health has been reduced to T'_L. On the other hand, the time he spends producing health reduces his time available for other activities. Time spent on health investment increases health stock and, in turn, reduces time lost to illness.

If the net effect of $T'_H + T'_L$ is a gain in available time, then this illustrates the pure investment aspect of health demand. The health investments "pay off" in terms that both add to potential leisure and also increase the potential income, shifting the income-leisure line outward from VS to RB. The expenditure of time for health producing activities may later improve Ed's available hours because he is less sick of productive activity.

As a result of his investment, Ed can increase his utility moving from point E to E'. Not only does investment in health lead to his feeling better, but it also leads to more future income and may lead to more leisure, as well. So, it is possible to earn more money and more leisure time together.

Ques-04:- According to Grossman model, how health care demand is different from tranditional approach of demand?

Answer:-
Grossman used the theory of human capital to explain the demand for health and health care. According to human capital theory, individuals invest in themselves through education, training, and heath to increase their earnings. Health investment Leads to a myraid of decisions regarding work, health care, exercise and the use of consumption goods and bads (such as unhealthy food, cigarettes, alcohol, or addictive drugs). In particular, according to this theory:

(i) It is not medical care as such that consumers want, but rather health. In seeking health, they demand medical care inputs to produce it.

(ii) Consumers do not merely purchase health passively from the market. Instead, they produce health, combining time devoted to health-improving efforts including diet and exercise with purchased medical inputs.

(iii) Health Lasts for more than a day or a month or a year. It does not depreciate instantly, and it can be analyzed like a capital good.

(iv) Perhaps most importantly, health can be treated both as a consumption good and an investment good. People desire health as a consumption good because it makes them feel and look better. As an investment good, they desire health because it increases the number of healthy days available to work and earn income.

These are the reasons of how health care demand is different from traditional approach of demand.

Ques-05:- How optimum health stock is determined? Explain with diagram

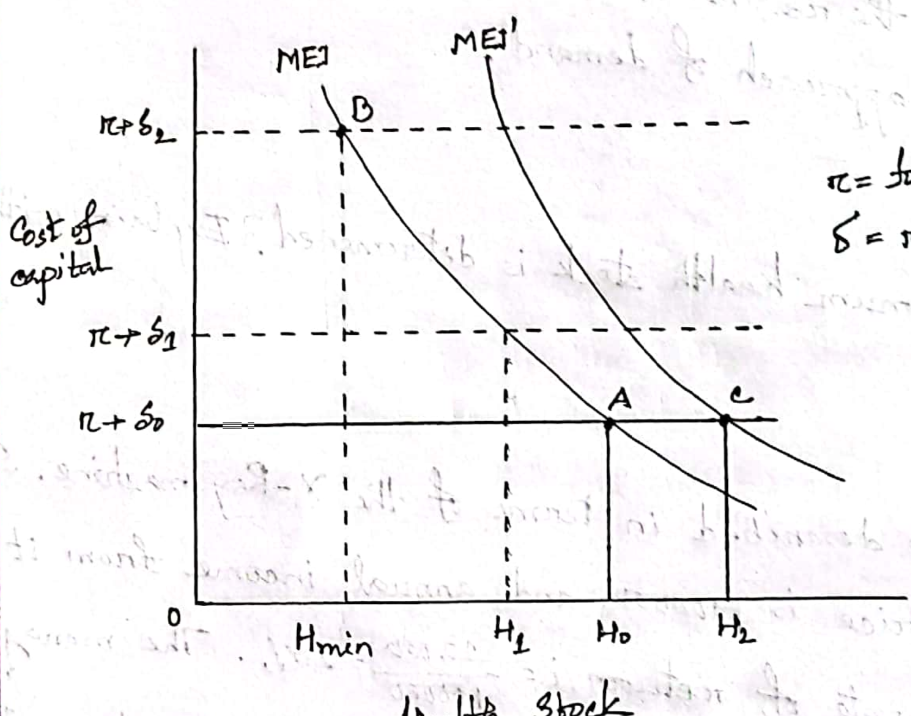
Answer:-

MEI:-

The MEI can be described in terms of the X-Ray machine. Suppose, a X-Ray machine price is $100000 and annual income from it is $20000, that the rate of return, $i = \frac{20,000}{100000} = 20\%$. The management would buy this machine if rate covers its opportunity cost and depreciation cost.

If management considered owning two machines, then the second machine would have a lower rate of return than the first. Because first machine would assign it to the highest priority uses, those with the highest rate of return. And second machine would assign it to the lesser priority uses (might be idle occasionally). The clinic will purchase the second machine only if its rate of return is still higher than the ~~rate of return~~ interest ~~rate~~ plus depriciation.

The Decreasing MEI:-



r = forgone interest (rate)
δ = rate of depreciation

Figure; Optimal Health Stock

Other machines probably could be added at successively lower rates of return. In the figure, the marginal efficiency of investment curve, MEI, describes the pattern of rates of return, declining as the amount of investment increases. The cost of capital (that is, the interest rate, r, plus the depreciation rate, δ_0) is shown as the horizontal line labeled $r+\delta_0$. The optimum amount of capital demanded is thus H_0, which represents the amount of capital at which the marginal efficiency of investment just equals the cost of capital. This equilibrium occurs at point A.

Like the marginal efficiency of investment curve in this example, the MEI curve for investment in health also would be downward sloping. This occurs because the production function for healthy days exhibits diminishing marginal returns. The cost of capital for health would similarly reflect the interest rate plus the rate of depreciation in health. Understand that a person's health, like any capital good, also will depreciate over time. As we age, certain joints may wear out, certain organs may function less well than before or we may become more forgetful. Thus, the optimal demand for health is likewise given at the intersection of the MEI curve and the cost of capital curve $(r+\delta_0)$.

Ques- 06: How does a person's investment in health change in response to change in age, wage and education?

Answer:-

Age:-

A person's health stock may depreciate faster during some periods of ~~time~~ life and more slowly than others. Eventually, as he ages, the depreciation rate will likely increase.

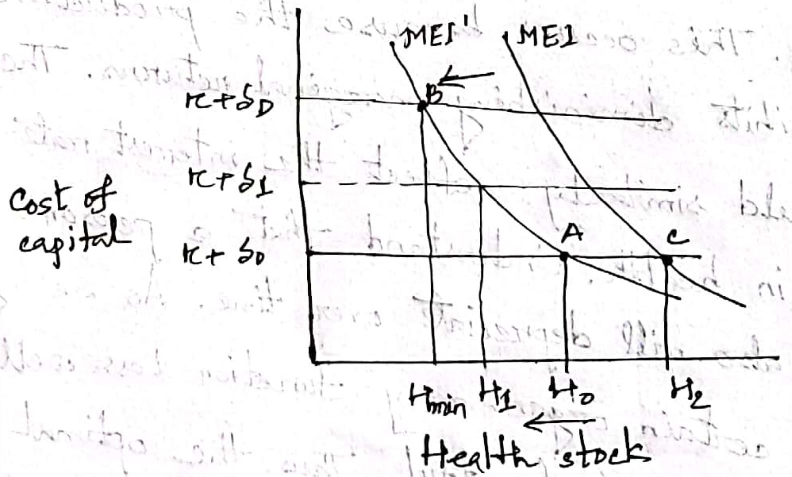


Fig-1: Effect of age in person's health stock

From the fig-1, we see by hypothesis, the depreciation rate, δ increases with age from δ_0 to δ_1 and ultimately δ_D. These assumptions imply that the optimal health stock decreases with age. The optimal health at younger age, H_0, is greater than H_1, the optimal health at older age. Higher depreciation rates increase the cost of holding capital stock. In old age, health depreciation rates are extremely high, δ_D, and optimal health stock fall to H_{min}, at point B. This suggests that as expected length of life decreases, the MEI curve shifts to the left, because the returns from an investment will last for a shorter period of time.

Wage Rate:-

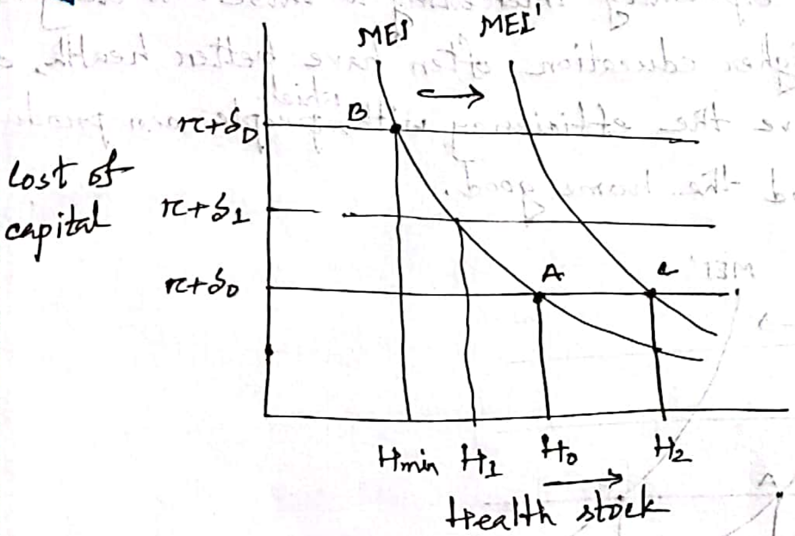


Fig-2: Effect of increasing wage rate

Fig-2 illustrates the effect of a change in the wage rate. Increased wage rates increase the returns obtained from healthy days. Thus, higher wages imply a higher MEI curve, or MEI'.

Assume now that the original MEI curve describes the lower-wage case and yeilds optimal health stock, H_0. The MEI' curve, above MEI, shows the marginal efficiency of investment for someone with higher wages. At new equilibrium point C, the higher wage will imply a higher ~~optimum~~ optimal level of health stock, H_2, in this pure investment model. The rewards of being healthy are greater for higher-wage workers, so increased wages will generally tend to increase the optimal capital stock.

Education:-

Education is especially interesting to those who study health demand. Those with higher education often have better health, and education may improve the efficiency with which people can produce investments to health and the home good.

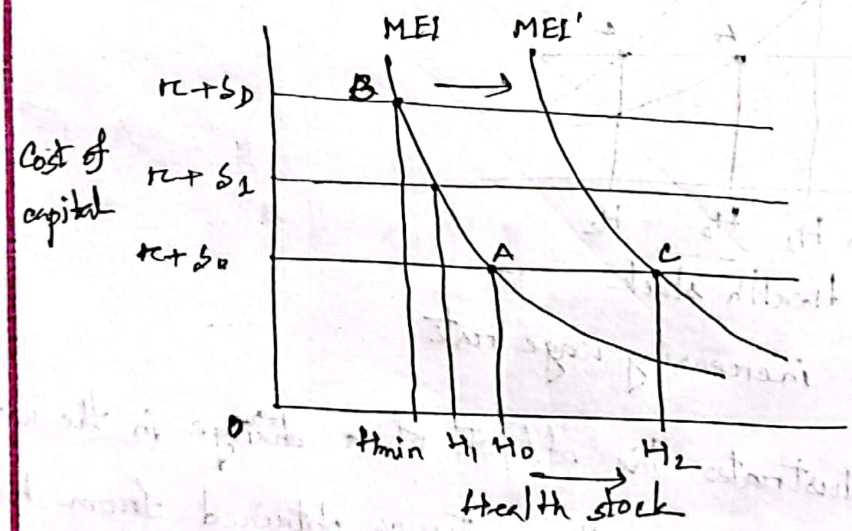


Fig-3: Effect of education

Fig-3 illustrates the effect of education. Here, MEI curve illustrates the marginal efficiency of investment for the consumer with a low level of education, while the MEI' curve illustrates the same person with a higher level of education. This model indicates that because education raises the marginal product of the direct inputs, it reduces the quantity of these inputs required to produce a given amount of gross investment.

It follows that given investments in health can be generated at less cost for educated people, and thus they experience higher rates of return to a given stock of ~~capital~~ health. The result, as shown, is that the more highly educated people will choose higher optimal health stocks, H_2, the less highly educated will choose H_0.

Ques:- Using supply and demand graph and assuming competitive markets, show and explain the effect ~~in~~ equilibrium price and quantity of the following:-

(a) A decrease of the price of Ace on the market for Napa. (Hint: Ace and Napa are substitutes).

Answer:-

Since Ace and Napa are substitute goods, a decrease of the price of Ace will decrease the demand for Napa.

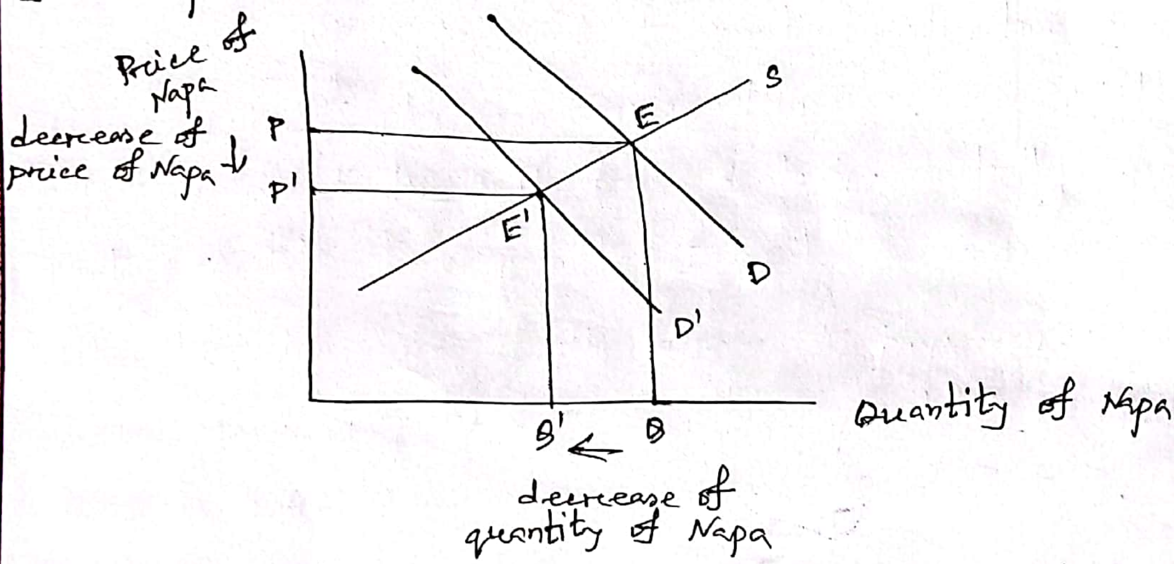


Fig-1: Effect of decreased price of Ace and market for Napa

As demand for Napa decreases, the demand curve shifts to the left from D to D'. Therefore, the quantity of Napa decreases from Q to Q'. The equilibrium position of the competitive market shifted from E to E' and price decreases from P to P'. These are shown in fig-1.

(b) Increased price of painkillers on the market for antiulcerative drugs. (Hint: Painkillers and antiulcerative drugs are complements).

Answer:-

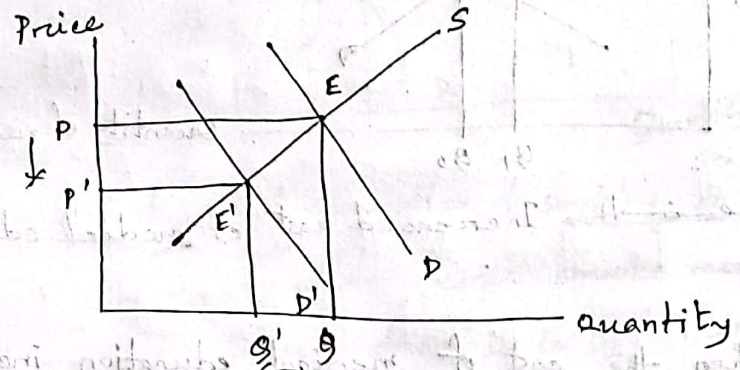


Fig-2: Effect of increased price of painkillers and market for antiulcerative drugs.

Painkillers and antiulcerative drugs are complementary goods. When prices of painkillers increases, then the demand for painkillers decreases. This will decrease the demand and quantity of the antiulcerative drugs in the market. Therefore, equilibrium shifts from E to E' and price decreases from P to P'.

(c) Increased cost of medical education on the market for physician services.

Answer:-
If the cost of medical education increases, it will lead to decrease in the supply of physical services. Because of high cost of

medical education, high price of medical services is charged.

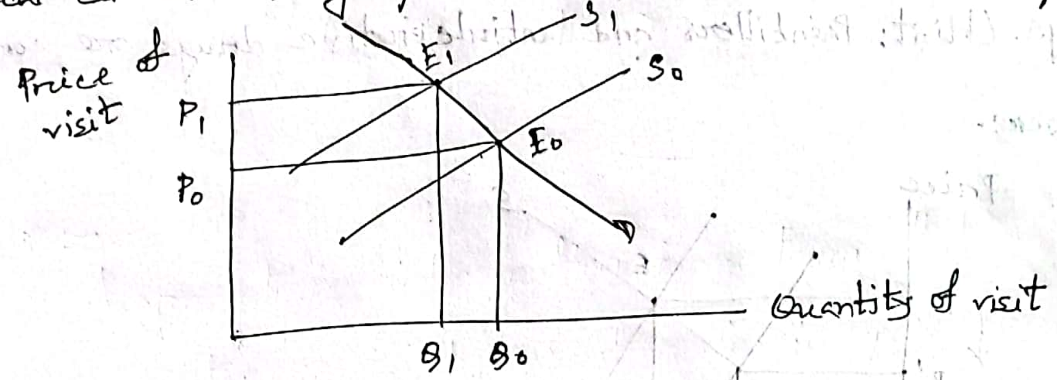


Fig-3: ~~Decrease in the~~ Increased cost of medical education

When the cost of medical education increases, the supply of physician service decreases. This shifts the supply curve to the left from S_0 to S_1. Initially equilibrium point $E(Q_0, P_0)$, where quantity of visit was Q_0 and price was P_0. After the decrease of the supply of physician services, there creates ~~an~~ new equilibrium E' where quantity of visit decreases from Q_0 to Q_1 and price increases from P_0 to P_1.

(d) The virtual elimination of smoking on the market for hospital services.

Answer:-

When more people engage in smoking, then demand for hospital services will increase. Therefore, its cost will increase.

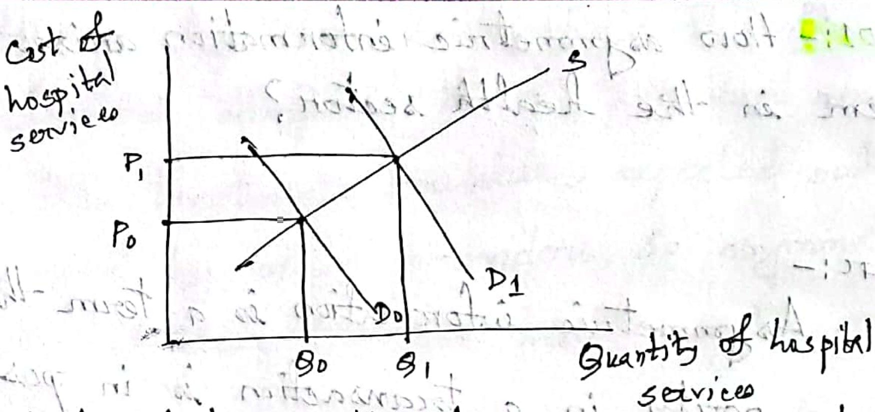


Fig-4:- Effect of virtual elimination of smoking on the market for hospital services.

In fig-4, when people smoke more then demand for hospital services increases from D_0 to D_1. Therefore price of medical services increases from P_0 to P_1. Thus new equilibrium for hospital services creates from E_0 to E_1.

Chapter-10

Ques-01:- How asymmetric information arises inefficiencies problem in the health sector?

Answer:-

Asymmetric information is a term that refers to when one party in a transaction is in possession of more information than the other. In the health sector, asymmetric information is when a patient is less well informed about appropriate treatments than the attending physician. The market for many health care services and for insurance in particular exhibit significant degrees of asymmetric information and agency relationships. For example, adverse selection, a phenomenon in which insurance attracts patients who are likely to use services at a higher than average rate, results from asymmetric information. Potential beneficiaries have better information than the insurer about their health status and their expected degree of health care. As a result, premiums for high-risk patients will be underpriced, whereas the opposite holds true for lower-risk patients.

Level of information will differ among participants, such as between physicians and patients. Patients are often poorly informed compared to the provider about theire conditions, the treatments available, expected outcomes, and prices charged by other providers. So asymmetric information reduces the efficiency in the health sector.

According to Pauly (1978), one-fourth or more of total personal health care expenditure might be regarded as 'reasonably informed'. If we add nursing home services and chronic disease conditions, ~~this~~ ratio comes to one-third.

Ques-02:- Explain ~~the~~ asymmetric information using lemons principle.

Answer:-

Nobel Laureate George Akerlof introduced the idea of asymmetric information, through an analysis of the used-car market. Suppose that nine used cars are to be sold (potentially) that vary in quality from 0, meaning a 'lemon', to a high of 2, meaning a 'cream puff'. Suppose that the nine cars have respectively quality levels (Q) given by the cardinal index values of

$0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1\frac{1}{4}, 1\frac{1}{2}, 1\frac{3}{4}$, and 2. A car with a value of 1 has ~~two~~ twice the quality of a car with an index of $\frac{1}{2}$.

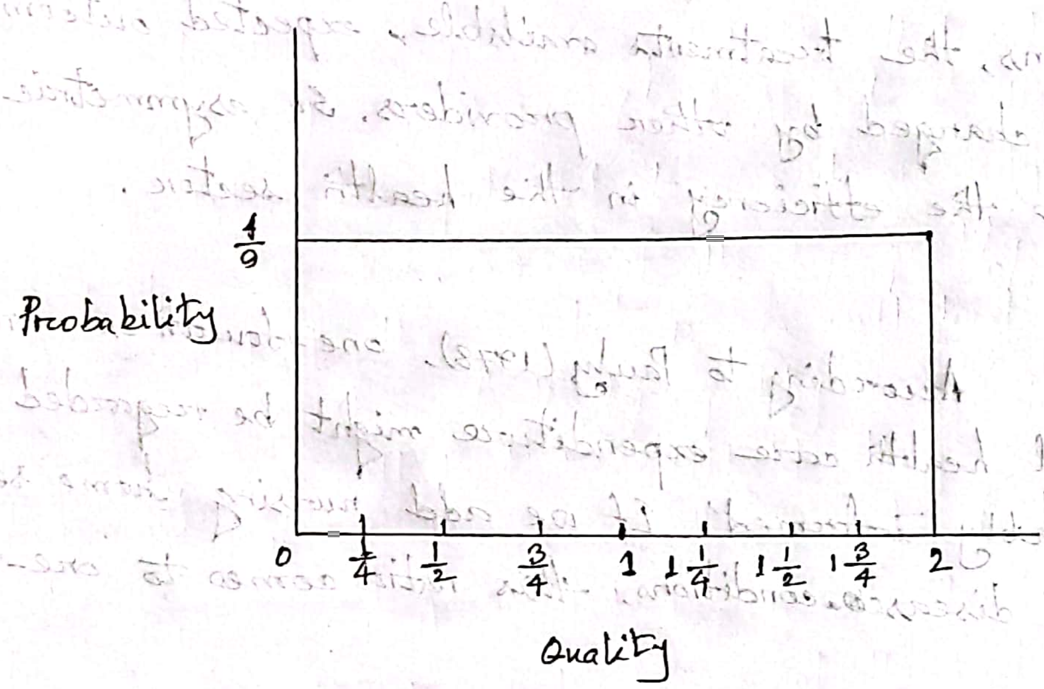


Figure: The availability of products of different quality

The horizontal axis shows the quality level and the vertical axis shows the uniform probability.

Suppose a car owner (and potential seller) knows its quality level exactly but that potential buyers know only the distribution of quality. Reverse value to the seller = $5500 × Q. That is, the owners would sell their cars only if they could get at least $5000 for every unit of car quality. In contrast, the nonowners value them $7500 × Q. Sales take place when the auctioneer finds a price that successfully equates quantity demanded with quantity supplied.

If the auctioneer calls out an initial price of \$10000 per car, all owners know it is worth while to sell their cars, so all nine cars will be supplied. However, nonowners would not buy any cars at a price of \$10000 because they are willing to pay only \$7500 per unit of quality. They guess that all cars have a quality of 1, for a product of \$7500 × 1 = \$7500. They would be willing to buy cars only if the price were less than or equal to \$7500.

So the auctioneer tries a lower price, say \$7500. Unfortunately, at this price, the owners of the two best cars will withdrawn from the market. The owner of the car with two units of quality receiving only \$7500 ÷ 2, or \$3750 per unit of quality; the owner of the the car with $1\frac{3}{4}$ units will act the same way. The withdrawal of the two best cars causes the average of remaining seven cars to fall. Now at a price of \$7500 per car, the best quality car is $1\frac{1}{2}$ unit, and the average quality will be $\frac{3}{4}$. Potential buyers are now willing to pay \$7500 × $\frac{3}{4}$ or, \$5625 for any car. The new price of $7500 is too high for buyers.

In this example no equilibrium that satisfies both buyers and sellers will be found. Akerlof saw the problem this way: When ~~the~~ potential buyers know only the average quality of used cars, ~~Owners of the top-quality cars will tend to withhold~~ then market prices will tend to be lower than the true value of the top-quality cars. Owners of the top-quality cars will tend to withhold their cars from sale. In a sense, the good cars are driven out of the market by the lemons. Under what has become known as the Lemons principle, the bad drives out the good until no market is left.

Ques-03:- Show the application of Lemons Principle in health insurance market.

Answer:-

We can apply the Lemons Principle directly to health insurance market. In fig-2, let the horizontal axis measure the expected health expenditure levels of a population of n potentially insured people.

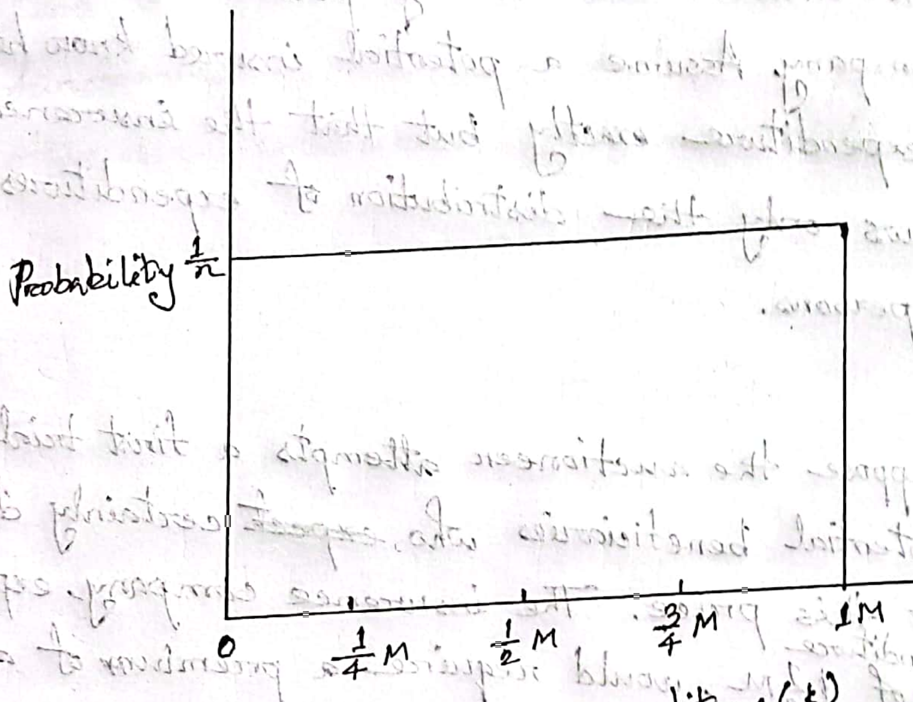


Fig-2: Uniform probability of expenditure

Assume that their expected health expenditure levels for the insured period range from a low of $0 up to an expenditure level of $1M. The vertical axis represents the probability with a uniform distribution so that the probability of any level of spending is $\frac{1}{n}$. The premium (or price) received from each buyer must cover the insured population's average expenditure and other expenses.

Information asymmetry will likely to occur because the potential insureds know more about their expected health expenditures in the coming period than does the insurance company. Assume a potential insured know his or her future expenditure exactly but that the insurance company knows only the distribution of expenditures for all insured persons.

Suppose the auctioneer attempts a first trial price of $\$0$. All potential beneficiaries who ~~expect~~ certainly demand coverage at this price. The insurance company, expecting an average expenditure of $\$\frac{1}{2}M$, would require a premium of at least $\$\frac{1}{2}M$.

Following Akerlof analysis, suppose the auctioneer tries a higher price, say $\$\frac{1}{2}M$. In this case, all potential beneficiaries who expect an expenditure level below $\$\frac{1}{2}M$ will choose to self-insure, that is, leave the insurance market altogether, because this premium is higher than their privately known levels of health expenditure. When these healthier people leave the market, the average expected expenditure level of the remaining insured persons rises from $\$\frac{1}{2}$ to $\$\frac{3}{4}$M. Thus, the higher ~~risk~~ health risks tend to drive out the lower health risk people, and a functioning market may fail to appear at all for some otherwise - insurable health care risks.